

6. ASSIGNMENT PROBLEM

NBS

NUMERICAL QUESTIONS

1. 2070 Old Q.No. 9a

A company has 4 different types of works to be performed in 4 different machines. Each type of work must be done on one and only one machine. Give the optimum assignments so that the cost is minimum. [10]

Machines	Works			
	W ₁	W ₂	W ₃	W ₄
M ₁	12	18	44	23
M ₂	30	33	25	30
M ₃	21	9	24	28
M ₄	15	31	21	14

Ans: Rs. 60

2. 2069 (Old) Q. No. 8b

Five different jobs are to be assigned to four different machines so as to minimize the cost. Find the optimal assignment with the corresponding cost.

Jobs	Machines			
	M ₁	M ₂	M ₃	M ₄
J ₁	90	70	60	20
J ₂	60	60	70	60
J ₃	50	30	40	40
J ₄	40	20	50	90
J ₅	20	80	30	60

Ans: Total minimum cost = Rs. 100

3. 2068 Q.No. 9

A project work consists of four major jobs for which an equal number of contractors have submitted tenders. The tender amount quoted (in millions of rupees) are as follows:

Contractors	Jobs			
	P	Q	R	S
W	30	44	50	35
X	36	42	48	32
Y	32	40	52	30
Z	29	46	54	36

Find minimum cost of optimal assignment, when each contractor has to be assigned at least one job. [10]

Ans: Total min cost = Rs 151

4. 2068 Old Q.No. 8b

Four different transportation companies are assigned to handle three cargo consignments with a view to maximize the profit. Find optimal assignment so that the profit can be maximized. [10]

Transportation companies	Cargo consignment		
	C ₁	C ₂	C ₃
T ₁	50	200	250
T ₂	100	150	150
T ₃	150	150	150
T ₄	250	50	100

Ans: Total max. profit = Rs.650

5. 2067 Q.No. 8b (Old)

Assign jobs to different machines in order to minimize the cost from the information given below and compute minimum cost.

Machines	Job		
	P	Q	R
A	18	8	10
B	24	13	15
C	28	17	19
D	32	19	22

Ans: Total Min. cost = Rs. 50; A - P; B - Q; C - R; D - S

6. 2066 Q.No. 8 b

A department has four subordinates and four tasks to be performed. The subordinates differ in efficiency and the tasks differ in their intrinsic difficulty. The estimates of the profit in rupees each man would earn is given in the effectiveness matrix below. How should the tasks be allocated, one to each man, so as to maximize the total earnings?

(Subordinates)	Tasks			
	1	2	3	4
1	5	40	20	5
2	25	35	30	25
3	15	25	20	10
4	15	5	30	15

Ans: Task 1 = man 3; Task 2 = man 1; Task 3 = man 4; Task 4 = man 2; Total cost = Rs. 110

7. 2063 Q.No. 8 (a)

From the following find the optimal assignment so that the profit is maximized on the basis of one machine to one job.

Profit in 'Rs.' Jobs				
Machines	A	B	C	D
X	1800	2400	2800	3200
Y	800	1300	1700	1900
Z	1000	1500	1900	2200

[10]

8. 2062 Q.No. 8 (b)

Ans: Maximum profit = Rs. 6,400

Four children in a household were assigned three different household chores to be done. The children are motivated to get pocket money for the job. Assign the jobs to the children in such a way that their pocket money income is maximum.

[10]

Children	Clean the house	Wash clothes	Cook dinner
Ram	1	4	5
Laxman	2	3	3
Bharat	3	3	3
Shatrughan	5	1	2

9. 2059 Q.No. 8 (a)

Ans: Maximum pocket money income = Rs. 13

The ABC Company has three jobs to be done on three machines. Each job must be done on one and only one machine. The cost of each job on each machine is given in the following table.

Jobs	Machine		
	X	Y	Z
A	4	6	8
B	2	3	4
C	4	8	5

Give the job assignments, which will minimize cost.

Ans: Minimum cost = Rs. 12; X → A, Y → B, Z → C

NBA**10. 2056 Q.No 5 (a)**

Assign the job to machine in the following data of opportunity cost to minimize opportunity cost:

Jobs	Machine		
	M ₁	M ₂	M ₃
A	Rs. 25	Rs. 31	Rs. 35
B	Rs. 15	Rs. 20	Rs. 24
C	Rs. 22	Rs. 19	Rs. 17

Ans: Minimum cost = Rs. 62; A→X, B→Y, C→Z

11. 2054 Q.No. 7 (a)

The innovative product company is considering an expansion into five new sales districts. The company has been able to hire four new experienced sales persons. Upon analysing the new sales person's experience in combination with a personality test which was given to them, the company assigned a rating to each of the salespersons for each of the districts. These ratings are as follows:

Sales Persons	Districts				
	1	2	3	4	5
A	92	90	94	91	83
B	84	88	96	82	81
C	90	90	93	86	93
D	78	94	89	84	88

The company knows that with four sales persons, only four of the five potential districts can be covered. It would like to know which four districts the sales persons should be assigned in order to maximize the total of the ratings.

Ans: Maximum total ratings = Rs. 375

12. 2048 Q.No. 6 (a)

Find the optimal assignment for the given problem, where four person are assigned to four jobs.

Persons	Jobs			
	1	2	3	4
1	3	5	11	4
2	2	1	6	9
3	8	10	2	7
4	10	6	1	5

Ans: Minimum assignment cost = Rs. 11

13. 2046 Q.No. 6 (a)

The advanced company has three jobs to be done on three machines. Each job must be done on one and only one machine. The cost of each job on each machine is given in the following table.

Cost information:

Jobs	Machine		
	X	Y	Z
A	Rs. 25	Rs. 31	Rs. 35
B	Rs. 15	Rs. 20	Rs. 24
C	Rs. 22	Rs. 19	Rs. 17

Give the job assignments, which will minimize the cost.

Ans: Minimum cost = Rs. 62; A→X, B→Y, C→Z

14. 2041 Q.No. 6 (a)

The ABC Motor works has four machines on which to do four jobs. Each job can be assigned only one machine. The cost of each job on each machine is given in the following table. What are the job assignments which will minimize costs?

Machines	Jobs			
	1	2	3	4
W	3	5	11	4
X	2	1	6	9
Y	8	10	2	7
Z	10	6	1	5

Ans: Minimum cost = Rs. 11